

WHYALLA AP, Australia

WMO: 956640

Lat: 33.0539S

Lon: 137.5206E

Elev: 7

StdP: 101.24

Time Zone: 9.50 (AUA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.7	3.1	-8.2	1.9	27.0	-5.3	2.4	24.7	12.4	17.3	11.3	17.0	2.6	320	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.3	39.4	18.9	36.8	18.5	34.0	17.9	22.6	28.6	21.7	28.3	20.9	27.8	7.0	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.2	15.9	24.0	19.9	14.7	24.0	18.9	13.7	23.4	66.9	28.2	63.6	28.5	60.3	27.7	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.7	10.6	9.7	-0.8	44.0	1.1	1.8	-1.6	45.3	-2.2	46.3	-2.8	47.3	-3.6	48.6		
			-1.2	24.3	1.2	1.4	-2.0	25.3	-2.8	26.1	-3.4	26.9	-4.3	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	23.9	23.7	21.3	18.0	14.4	11.8	11.1	12.0	14.7	17.3	20.0	21.9
	DBStd	5.58	3.90	3.67	3.38	3.08	2.61	2.41	2.36	2.87	3.50	3.91	3.97	3.88
	HDD10.0	34	0	0	0	0	1	10	14	9	1	0	0	0
	HDD18.3	1014	1	1	7	42	126	198	223	197	122	68	23	7
	CDD10.0	2763	431	383	349	240	137	62	49	72	142	228	301	368
	CDD18.3	701	174	151	98	32	4	0	0	2	13	37	74	117
	CDH23.3	7556	1739	1456	914	369	52	2	1	36	233	588	901	1266
	CDH26.7	3752	936	735	427	143	9	0	0	6	81	281	463	672
Wind	WSAvg	4.9	6.1	5.8	5.0	4.2	3.9	3.9	4.1	4.4	4.8	5.3	5.6	5.7
Precipitation	PrecAvg	249	12	14	17	20	27	28	24	25	33	26	20	18
	PrecMax	363	82	64	80	83	56	72	73	51	106	59	48	66
	PrecMin	147	0	0	0	1	6	4	6	1	1	3	0	1
	PrecStd	62	18	15	21	21	13	17	16	17	29	17	13	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.1	41.2	38.7	34.9	27.9	22.2	22.5	27.1	32.6	36.6	39.8	41.1
		MCWB	19.7	20.0	17.9	17.5	14.9	13.8	12.3	13.5	15.2	16.3	18.1	18.9
	2%	DB	39.6	37.9	34.7	30.6	24.5	19.7	20.2	23.7	29.0	33.1	35.5	37.2
		MCWB	19.3	19.6	18.2	16.0	14.2	12.6	11.5	12.2	13.9	15.3	16.8	18.1
	5%	DB	35.8	34.2	31.4	27.6	22.1	18.1	18.3	21.0	25.9	29.9	32.1	33.8
		MCWB	19.2	19.4	18.1	15.4	13.7	12.0	11.1	11.5	13.0	14.5	16.4	17.7
	10%	DB	31.6	31.1	28.1	24.7	20.1	16.8	16.8	18.5	22.8	26.1	28.4	30.0
		MCWB	18.9	19.2	18.0	15.4	13.4	11.7	10.8	11.0	12.6	13.9	16.4	17.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.7	25.0	22.9	19.9	17.5	16.2	14.3	15.0	17.0	18.6	21.1	22.3
		MCDB	29.8	28.2	27.5	23.3	20.8	18.8	18.0	23.1	25.7	26.8	27.6	28.3
	2%	WB	22.3	22.9	21.5	18.7	16.1	14.3	13.1	13.7	15.6	17.3	19.7	20.9
		MCDB	29.2	28.7	27.0	23.2	20.0	16.9	16.8	19.2	23.5	24.9	27.6	29.2
	5%	WB	21.5	21.9	20.4	17.8	15.3	13.4	12.3	12.8	14.6	16.4	18.7	19.8
		MCDB	28.7	28.2	26.0	23.0	19.3	16.2	16.1	17.6	21.1	23.7	26.5	27.9
	10%	WB	20.6	20.9	19.4	16.9	14.4	12.5	11.6	12.0	13.7	15.5	17.7	18.9
		MCDB	28.0	27.5	25.3	22.2	18.6	15.5	15.4	16.6	19.7	22.8	25.1	26.4
Mean Daily Temperature Range	5% DB	MDBR	11.3	10.7	10.6	11.2	10.5	9.9	11.0	11.8	12.6	12.3	11.5	11.4
		MCDBR	19.7	17.9	17.2	17.3	14.5	11.8	13.5	16.2	19.0	20.4	19.9	19.9
		MCWBR	6.0	5.6	6.4	7.4	6.8	6.3	7.0	7.6	7.5	7.5	6.5	6.2
	5% WB	MCDBR	13.1	11.7	12.1	11.7	10.7	9.3	11.0	12.6	14.2	14.9	14.2	13.7
		MCWBR	5.2	4.8	5.5	6.3	6.1	6.1	6.6	7.0	6.7	6.5	5.4	5.5
Clear-Sky Solar Irradiance	taub	0.348	0.348	0.328	0.320	0.308	0.298	0.298	0.308	0.335	0.344	0.350	0.341	
	taud	2.536	2.562	2.602	2.605	2.616	2.624	2.619	2.573	2.491	2.476	2.497	2.542	
	Ebn at Noon	990	970	956	910	870	855	873	909	931	963	982	1001	
	Edh at Noon	110	104	93	83	73	68	71	84	102	112	114	110	
All-Sky Solar Radiation	RadAvg	8.03	7.18	5.96	4.47	3.23	2.68	2.92	3.89	5.27	6.55	7.44	7.97	
	RadStd	0.39	0.37	0.31	0.21	0.13	0.15	0.13	0.21	0.34	0.33	0.33	0.35	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.23	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	+0.29	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page